

Lecture 08 - Exercises

These exercises accompany Lecture 08, “Hypothesis Testing.” They practice the logic of testing a benchmark claim: formulating hypotheses, choosing the direction of a test, computing standard errors and test statistics, using critical values and p-values, and writing careful conclusions.

The exercises are separated into two parts. The first part should be done by hand. This is important because manual work helps you understand the mechanics: what the benchmark is, how the hypotheses are written, how the standard error enters the calculation, how the rejection region is chosen, and what the decision actually means. Only after that does it become useful and convenient to move to larger datasets and use Python.

The second part uses Python. These exercises apply the same ideas to financial return data. Python is used to compute returns, test average returns against benchmarks, compare p-values with significance levels, and write short statistical conclusions.

Solutions are not included here. The by-hand solutions are provided separately in Markdown, and the Python solutions are provided in a notebook.

Part A - By-Hand Practice

Exercise 1 - Benchmark, Null, and Alternative

For each situation, identify:

1. the parameter of interest;
2. the benchmark claim;
3. the null hypothesis;
4. the alternative hypothesis.

Situation	Claim or question
A	A call center claims that average waiting time is at most 90 seconds.
B	A trading strategy is useful only if its expected monthly excess return is positive.
C	A quality-control team wants to know whether average ring diameter differs from 5 cm.
D	A bank is concerned that the average time to process a loan application is above 5 business days.

Exercise 2 - One-Sided or Two-Sided?

For each situation, say whether the test should be one-sided or two-sided. Explain briefly.

1. A delivery company wants evidence that average delivery time is below 3 days.
2. A manufacturer wants to know whether a machine has moved away from a target diameter of 5 cm.
3. A media team will keep a redesigned homepage only if the average time users spend on it is higher than the current 40 seconds.
4. A finance team wants to know whether the expected return of a strategy is different from zero.

5. A risk team is concerned only if average loss exceeds a warning threshold.

Exercise 3 - Type I and Type II Errors

A call center claims that average waiting time is at most 90 seconds. A client tests this claim against the alternative that the average waiting time is above 90 seconds.

1. State H_0 and H_1 .
2. In this context, what would a Type I error mean?
3. In this context, what would a Type II error mean?
4. Which error may be more costly for the client? Explain briefly.

Exercise 4 - Standard Error and Test Statistic

A sample of 100 calls gives a mean waiting time of 93 seconds. Treat the population standard deviation as 10 seconds. The benchmark mean is 90 seconds.

1. Compute the standard error of the sample mean.
2. Compute the z-statistic.
3. Interpret the z-statistic in words.
4. Why is the raw difference $93 - 90 = 3$ not enough by itself?

Exercise 5 - Right-Tail Critical Value Decision

A right-tail test uses:

$$H_0 : \mu \leq 90, \quad H_1 : \mu > 90.$$

The test statistic is $z = 3$.

1. At the 5% significance level, use critical value 1.645. Reject or fail to reject?
2. At the 1% significance level, use critical value 2.326. Reject or fail to reject?
3. Write a careful conclusion at the 1% level.

Exercise 6 - Left-Tail Test

A delivery company claims that average delivery time has been reduced below 3 days. A random sample gives:

$$\bar{x} = 2.82, \quad \sigma = 0.60, \quad n = 100.$$

Test the claim using:

$$H_0 : \mu \geq 3, \quad H_1 : \mu < 3.$$

1. Is this a right-tail, left-tail, or two-sided test?
2. Compute the standard error.
3. Compute the z-statistic.
4. At the 5% level, use critical value -1.645. Reject or fail to reject?
5. Write the conclusion in context.

Exercise 7 - Two-Sided Quality-Control Test

A machine should produce rings with average diameter 5 cm. A sample of 64 rings gives:

$$\bar{x} = 5.08, \quad \sigma = 0.24, \quad n = 64.$$

Test whether the process has moved away from the target:

$$H_0 : \mu = 5, \quad H_1 : \mu \neq 5.$$

1. Compute the standard error.
2. Compute the z-statistic.
3. At the 5% level, use critical values -1.96 and 1.96. Reject or fail to reject?
4. Write the conclusion in context.

Exercise 8 - P-Value Decisions

For each p-value, state the decision at the 5% significance level.

Case	p-value	Decision at 5%
A	0.003	
B	0.028	
C	0.071	
D	0.410	

Then explain why “fail to reject” does not mean “prove the null hypothesis is true.”

Exercise 9 - Interpreting a P-Value

A one-sided test of

$$H_0 : \mu = 0$$

against

$$H_1 : \mu > 0$$

returns a p-value of 0.08.

1. Interpret the p-value correctly.
2. Is the result statistically significant at the 5% level?
3. Is the result statistically significant at the 10% level?
4. Give one incorrect interpretation of the p-value and explain why it is wrong.

Exercise 10 - Confidence Interval and Test Logic

A company estimates average delivery time as 2.8 days. An approximate 95% confidence interval for the unknown average delivery time is:

$$[2.5, 3.1].$$

The operational target is an average delivery time below 3 days.

1. Is the point estimate below the target?
2. Does the interval include 3 days?
3. What does this imply about the strength of evidence that the true average delivery time is below 3 days?
4. Write a careful interpretation.

Exercise 11 - Statistical Significance Versus Practical Importance

A trading signal has:

- estimated average daily return: 0.003%;
 - p-value: 0.002.
1. Is the result statistically significant at the 5% level?
 2. Why might it still be economically unimportant?
 3. List at least three practical issues that should be checked before treating the signal as useful.

Exercise 12 - Critique a Bad Conclusion

A junior analyst writes:

The p-value is 0.03, so there is only a 3% probability that the null hypothesis is true. Since the result is significant, we have proved that the strategy works and should invest immediately.

Identify at least four problems with this conclusion.

Then rewrite the conclusion professionally.

Part B - Python Applications

Use the file:

Lecture 08 aapl daily 2012 2021.csv

The file contains Apple daily prices. The Python exercises compute daily returns from the adjusted closing price.

Exercise 13 - Load Prices and Compute Returns

1. Load the dataset.
2. Parse the date column and sort the data by date.
3. Compute daily simple returns from `Adj Close`.
4. Remove missing returns.

- Report the sample size, sample mean, sample standard deviation, and standard error of daily returns.

Exercise 14 - One-Sided Test for Positive Expected Return

Using the daily returns from Exercise 13, test whether the expected daily return is positive:

$$H_0 : \mu \leq 0, \quad H_1 : \mu > 0.$$

- Compute the z-statistic using the sample standard deviation as an estimate of volatility.
- Compute the one-sided p-value using the standard normal distribution.
- At the 5% significance level, reject or fail to reject?
- Write a careful conclusion.

Exercise 15 - Two-Sided Test Against Zero

Using the same returns, test:

$$H_0 : \mu = 0, \quad H_1 : \mu \neq 0.$$

- Compute the test statistic.
- Compute the two-sided p-value.
- At the 5% significance level, reject or fail to reject?
- Explain how the decision differs from the one-sided test, if it does.

Exercise 16 - Testing Against an Annual Return Benchmark

Suppose an investor wants evidence that expected annual return is above 10%. Use the simple approximation:

$$\mu_{daily,0} = 0.10/252.$$

- State the null and alternative hypotheses for the daily mean return.
- Compute the z-statistic for the benchmark 0.10/252.
- Compute the one-sided p-value.
- At the 5% significance level, reject or fail to reject?
- Write a careful conclusion.

Exercise 17 - Sensitivity to the Significance Level

For the one-sided test in Exercise 14:

- Compare the p-value with $\alpha = 10\%$, $\alpha = 5\%$, and $\alpha = 1\%$.
- Create a small table showing the decision at each significance level.
- Explain how changing α changes the decision rule.

Exercise 18 - Short Statistical Memo

Write a professional memo of about 150-200 words answering:

Does the Apple sample provide strong evidence that expected daily return was positive during this period?

The memo should include:

1. the sample period and variable;
2. the hypotheses;
3. the sample mean and standard error;
4. the p-value;
5. the decision at the chosen significance level;
6. one practical caution about interpreting the result.